

《热力学与传热学基础》大作业
传热学部分

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2. 高速对流换热问题

在风洞中放置一块长为1.2 m，宽为1 m的平板，主流流动为 $Ma_\infty = 3$ ， $p_\infty = 1/20 \text{ atm}$ ， $T_\infty = -N^\circ\text{C}$ 。为了使平板温度维持在 30°C ，应当需要多大的冷却量？(N为学号后两位)

超声速主流滞止温度、静温关系式为 $\frac{T_0}{T_\infty} = 1 + \frac{\gamma-1}{2} Ma_\infty^2$ ，式中 T_0 为滞止温度、 T_∞ 为主流静温， Ma_∞ 为主流马赫数；对于实际边界层流动，由于粘性不可逆作用，并非全部动能转换为热能，定义恢复系数 $r = \frac{T_{aw} - T_\infty}{T_0 - T_\infty}$ ，式中 T_{aw} 为绝热壁温（恢复温度）；

由于高速边界层中显著的温度梯度，流体选取参考温度 T^* 时的常物性进行计算，参考温度表达式为

$$T^* = T_\infty + 0.50(T_w - T_\infty) + 0.22(T_{aw} - T_\infty) \quad p^* = p_\infty \quad \text{所需物性请用REFPROP进行查询。}$$

流体为空气可视作理想气体， $\gamma=1.4$ ， $R_g=287 \text{ J/(kg}\cdot\text{K)}$ ，标准状态下空气密度为 1.29 kg/m^3

$$\text{已知层流: } r = \text{Pr}^{*1/2} \quad \text{Nu}_x^* = 0.332 \text{Re}_x^{*1/2} \text{Pr}^{*1/3}$$

$$\text{湍流: } r = \text{Pr}^{*1/3} \quad \text{Nu}_x^* = 0.0296 \text{Re}_x^{*4/5} \text{Pr}^{*1/3}$$

Re^* 、 Pr^* 、 Nu^* 按参考温度 T^* 时的物性计算

解：

学号后两位54，主流静温

$$T_\infty = -54^\circ\text{C} = 219.15\text{K}$$

滞止温度：

$$T_0 = T_\infty \left(1 + \frac{\gamma-1}{2} Ma_\infty^2 \right) = 219.15 \left(1 + \frac{1.4-1}{2} \cdot 3^2 \right) = 613.62\text{K} = 340.47^\circ\text{C}$$

主流静压：

$$p_\infty = \frac{1}{20} \text{ atm} = \frac{1}{20} \times 1.01325 \times 10^5 \text{ Pa} = 5.06625 \text{ kPa} = p^*$$

恢复系数：

$$r = \frac{T_{aw} - T_\infty}{T_0 - T_\infty} \Rightarrow T_{aw} = r \cdot (T_0 - T_\infty) + T_\infty$$

主流速度：

$$u_\infty = Ma_\infty \sqrt{\gamma R_g T_\infty} = 3 \cdot \sqrt{1.4 \cdot 287 \cdot 219.15} = 890.2192 \text{ m/s}$$

首先，调用 Matlab 的 REFPROP 查询功能可得

$$T_{\infty} = 219.15K, \quad \nu_{\infty} = 1.78543469 \times 10^{-4} m^2/s$$

我们必须先计算出转换点：

$$Re_{cr} = \frac{u_{\infty} x_{cr}}{\nu_{\infty}} = \frac{890.2192 \cdot x_{cr}}{1.7854369 \times 10^{-4}} = 5 \times 10^5 \Rightarrow x_{cr} = 0.100281m$$

这里先计算湍流段：

$$r = Pr^{1/3}, \quad Nu_x = 0.0296 Re_x^{*4/5} Pr^{*1/3}$$

假设 $Pr = 0.71 \Rightarrow r = 0.71^{1/3} = 0.8921$

$$\Rightarrow T_{aw} = r \cdot (T_0 - T_{\infty}) + T_{\infty} = 0.8921 \cdot (613.62 - 219.15) + 219.15 = 571.0614K$$

$$\Rightarrow T^* = T_{\infty} + 0.50(T_w - T_{\infty}) + 0.22(T_{aw} - T_{\infty})$$

$$\Rightarrow = 219.15 + 0.5(30 + 273.15 - 219.15) + 0.22(571.0614 - 219.15)$$

$$\Rightarrow = 338.5705K = 65.4205^{\circ}C$$

调用 REFPROP 查询可得

$$Pr^* = 0.702281 \neq 0.71$$

我们可以利用 matlab 通过迭代的方式来找出精确的 Pr 数。

这里假设普朗特数 $Pr = 0.1$, 结果如下图所示

```
%REFPROP Initialization
addpath( 'C:\Program Files (x86)\REFPROP' );
setenv('RPPREFIX', 'C:\Program Files (x86)\REFPROP');

% Initial Parameters
T_inf = 219.15;      % Main Stream Static Temperature (K)
T_0 = 613.62;       % Stagnation Temperature (K)
T_w = 303.15;       % Wall Temperature (K)
p_inf = 5.06625e3;   % Main Stream Static Pressure (Pa)
tol = 1e-8;         % Convergence Tolerance
Pr_guess = 0.1;     % Initial Guess for Prandtl Number
max_iter = 100;     % Maximum Iterations
fluid = 'air';      % Fluid Type

% Set Up Parameters
Pr_old = Pr_guess;
iter = 0;
converged = false;

while ~converged && iter < max_iter
    iter = iter + 1;

    % Calculate the Coefficient of Restitution r
    r = Pr_old^(1/3); % Turbulent Flow

    % Calculate the Temperature of Adiabatic Wall T_aw
    T_aw = T_inf + r * (T_0 - T_inf);
```

```

% Calculate Reference Temperature T*
T_star = T_inf + 0.5 * (T_w - T_inf) + 0.22 * (T_aw - T_inf);

% Get Physical Properties Parameters by using REFPROP
Pr_new = refpropm('^', 'T', T_star, 'P', p_inf / 1e3, fluid); % Get Pr Number

% Check for Convergence
if abs(Pr_new - Pr_old) < tol
    converged = true;
end

% Update Pr Number
Pr_old = Pr_new;
nu = refpropm('$', 'T', T_star, 'P', p_inf / 1e3, fluid); % Get \nu Kinematic Viscosity
lambda = refpropm('L', 'T', T_star, 'P', p_inf / 1e3, fluid); % Get lambda Thermal Conductivity

end

% Display Updated Pr Number
if converged
    fprintf('lambda* = %.4e W/(m.K)\n', lambda);
    fprintf('v* = %.4e cm^2/s\n', nu);
    fprintf('Pr* = %.8f\n', Pr_new);
    fprintf('Taw = %.4f K\n', T_aw);
    fprintf('T* = %.4f K\n', T_star);
    fprintf('Number of Iterations: %d\n\n', iter);
else
    fprintf('Diverges, gg \n');
end

>> FP
lambda* = 2.9146e-02 W/(m.K)
v* = 3.8952e+00 cm^2/s
Pr* = 0.70230543
Taw = 569.7856 K
T* = 338.2898 K
Number of Iterations: 5

```

可见，仅需 5 次迭代即可求得误差小于 1×10^{-8} 的普朗特数 Pr^* 的精确值。

$$Pr_{Tur}^* = 0.70230543, \lambda_{Tur}^* = 2.9146 \times 10^{-2} \text{ W/m} \cdot \text{K}, \nu_{Tur}^* = 3.8952 \times 10^{-4} \text{ m}^2/\text{s}$$

$$r_{Tur} = Pr_{Tur}^{*1/3} = 0.70230543^{1/3} = 0.888776957 \approx 0.8888$$

$$T_{aw(Tur)} = 569.7856 \text{ K} = 296.6356^\circ \text{C} \approx 296.64^\circ \text{C}$$

$$T_{Tur}^* = 338.2898 \text{ K} = 65.1398^\circ \text{C} \approx 65.14^\circ \text{C}$$

$$Re_{x(Tur)}^* = \frac{u_\infty L}{\nu_{Tur}^*} = \frac{890.2192 \cdot x}{3.8952 \times 10^{-4}} \quad (0.1003 \text{ m} \leq x \leq 1.2 \text{ m})$$

$$Re_{L(Tur)}^* = \frac{890.2192 \cdot 1.2}{3.8952 \times 10^{-4}} = 2.7425 \times 10^6$$

$$Nu_x^* = 0.0296 Re_x^{*4/5} Pr^{*1/3}$$

由外掠等壁温平板湍流换热：

$$Nu_{L(Tur)}^* = 0.037 \left(Re_{L(Tur)}^{*4/5} - Re_{cr}^{4/5} \right) \cdot Pr_{Tur}^{*1/3}$$

$$h_{Tur}^* = 0.037 \left(Re_{L(Tur)}^{*4/5} - Re_{cr}^{4/5} \right) \cdot Pr_{Tur}^{*\frac{1}{3}} \cdot \lambda_{Tur}^* \int_{x_{cr}}^L x dx = 72.0788 W/m^2 \cdot K$$

同样的，我们可以利用 Matlab 代码计算层流段

```
>> FP2
lambda* = 2.8831e-02 W/(m.K)
v* = 3.8065e+00 cm^2/s
Pr* = 0.70269582
Taw = 549.8222 K
T* = 333.8979 K
Number of Iterations: 5
```

$$Pr_{Lam}^* = 0.70269582, \lambda_{Lam}^* = 2.8831 \times 10^{-2} W/m \cdot K, v_{Lam}^* = 3.8065 \times 10^{-4} m^2/s$$

$$r_{Lam} = Pr_{Lam}^{*1/3} = 0.70269582^{1/3} = 0.8890423652 \approx 0.8890$$

$$T_{aw(Lam)} = 549.8222 K = 276.6722^\circ C \approx 276.67^\circ C$$

$$T_{Lam}^* = 333.8979 K = 60.7479^\circ C \approx 60.75^\circ C$$

$$Nu_{x(Lam)}^* = 0.332 Re_{x(Lam)}^* Pr_{Lam}^{*1/3}$$

由外掠等壁温平板层流换热：

$$Nu_{L(Lam)}^* = 2 Nu_{x(Lam)}^*$$

$$\Rightarrow h_{Lam}^* = \left(0.664 \cdot \left(\frac{u_\infty}{v_{Lam}^*} \right)^{\frac{1}{2}} \cdot Pr_{Lam}^{*\frac{1}{3}} \cdot \lambda_{Lam}^* \right) \cdot \int_0^{x_{cr}} x^{-\frac{1}{2}} dx = 16.4860 W/m^2 \cdot K$$

总平均表面换热系数

$$h^* = h_{Lam}^* + h_{Tur}^* = 16.4860 + 72.0788 = 88.5648 W/m^2 \cdot K$$

冷却量

$$Q = A [h_{Lam}^* (T_w - T_{aw(Lam)}) + h_{Tur}^* (T_w - T_{aw(Tur)})]$$

$$= 1.2 \cdot 1 [16.4860 (303.15 - 549.8222) + 72.0788 (303.15 - 569.7856)]$$

$$= -27942.49437 W$$

需要冷却量 Q 为 **27.9425 kW**